THIS IS AN EXAMPLE

Portions have been removed to preserve the integrity and originality of the work done by a group of students.

Home Health Care App

CIS 245 Final Project

Prepared by:

Frederico Freshman, Susie Sophomore, and Jason Junior

FSJ

Software Solutions



1. OPPORTUNITY

Our client wishes to provide their Care Providers and Clients with software that will allow them to calculate billing and authorize the charges for the Care Provider and Client respectively. Our client requests a software solution from us and has given us the ability to acquire hardware and issue training relevant to the process of using this software for business purposes.

2. REQUIREMENTS ANALYSIS (Susie Sophomore)

The requirements listed below will be gathered through a careful process involving conducting interviews and meetings with the members of the company's management team. In addition to this, to ensure that the needs and preferences of all stakeholders are duly considered, surveys will be conducted amongst the PCPs and the current clients of the company. You can see the full requirements documentation *here*.

2.1 GENERAL REQUIREMENTS

#	Requirement Title	Requirement Description	Type	Priority	Who to Talk to	How to Gather Requirement	Who this Affects
1	Standard Fee Task List	Full list of tasks included in the PCP standard fee	Employee/ Training	Must Have	Management	Interview with Management	Client
2	Additional Fee Task List	Full list of tasks not included in the PCP standard fee	Employee/ Training	Must Have	Management	Interview with Management	Client
3	Standard vs Non-Standard Tasks	Create a process to identify which tasks fall outside the standard tasks list	Employee/ Training	Must Have	Management	Interview with Management	Client
4	Correct Business Licenses & Permissions	Ensure company has required licenses and permissions to operate the business	Legal	Must Have	Management / Legal	Interview with Management / Legal	Company
5	Compliance with Employment Laws	Ensure company is complying with all employment laws	Legal	Must Have	Management / Legal	Interview with Management / Legal	Company
6	Employee Certifications	Ensure company's employees have the correct certifications	Legal	Must Have	Management / Legal	Interview with Management / Legal	Company / PCP
7	Hiring Criteria for PCPs	Establish hiring criteria for PCPs (background checks, experience, certifications, education, etc.)	Hiring	Must Have	Management	Interview with Management	Company / PCP
17	Required Hardware	Employees must use a tablet device with our software preinstalled on the device	Employee/ Training	Must Have	PCP & Management	Interview with Management	PCP / Client

2.2 ANALYSIS

The four requirements below are needed in the program and will be incorporated during the design and implementation process.

#	Requirement Title	Requirement Description	Needed in Program	How to Implement
1	Standard Fee Task List	Full list of tasks included in the PCP standard fee.	Yes	-Collect fee schedule from management -Create dictionary/list with tasks included in standard fee.
2	Additional Fee Task List	Full list of tasks not included in the PCP standard fee and will require an additional fee.	Yes	-Collect fee schedule from management -Turn fee schedule into a list & assign a fee for these additional tasks.
3				
4				

2.3 INTERVIEW QUESTIONS FOR MANAGEMENT

1. Can you walk me through your current billing process?
2. .
3. How do you currently identify which tasks fall outside the standard tasks list?
4. .
5. What is the standard fee ($ amount)?
6. .
7. How should the additional fee for tasks be billed? By task?
8. .
9. Are you compliant with all employment laws for your state?
10. .
11. Do you provide training to PCPs for all required tasks including personal care, light housekeeping, meal preparation, and transportation? How?
12. .
13. Can you explain the communication plan between PCPs and management and how it is implemented?
14. .
15. Can you explain the billing and invoicing system process and how it is currently carried out?
16. .
17. How are employee records, client records, care plans, and billing information stored and secured?
18. .

2.3 PCP SURVEY QUESTIONS

1. On a scale of 1-5, how satisfied are you with the tasks included in the standard fee task list?
2. .
3. Are there any tasks you believe should be added to the standard fee task list?
4. .
5. .

2.4 CLIENT SURVEY QUESTIONS

1. Were you informed about the additional fee for certain tasks before it is added to your care plan?
2. .
3. On a scale of 1-5, how satisfied are you with the initial client assessment process?
4. .
5. Additional comments

3. PLANNING (Susie Sophomore)

3.1 PROPOSED SOLUTION

We are developing a new home health care application where the client is billed based on the specific tasks completed by the PCP, with a standard fee applied to all tasks identified in a provided list of general tasks. Additional fees will be charged if the PCP provides assistance that falls outside the standard list. To ensure transparency and clarity for the client, our application will provide the client with a list of non-standard tasks and issue an invoice for the services rendered for the client to sign off on.

We will separate the different functionalities into separate sections:

1. Task list management for standard tasks (defining tasks and the standard fee)
2. Task list management for additional fee tasks (additional fee tasks and their corresponding fees)
3. Fee calculation (calculating the total fee based on standard fee and each additional task)
4. Billing and invoicing (generating invoice for client to sign off on)

It is imperative that the organization furnish every PCP with a designated company-owned tablet, pre-loaded with our software/application, enabling them to enter their tasks and obtain client approval. To ensure the safety and integrity of sensitive data, appropriate security measures must be implemented to secure the tablet. We are estimating this project to be completed in 6 months. We are planning to train the company's already employed PCPs to use the application.

We are recommending that management, together with their employees, determine which tasks should be considered standard vs. non-standard and the corresponding fees should there be any changes in the future.

3.1 IN SCOPE

1. Our role for this project encompasses the entire software development life cycle of a home health care application from start to launch of the application.
2. .
3. Training sessions will be conducted to prepare management and employees for the application's launch and how to use the application.

3.2 OUT OF SCOPE

1. Business must have correct licenses & permissions and if they do not, the business must obtain them.
2. .
3. Business must ensure employees have correct certifications.
4. .
5. Business is responsible for providing training for PCPs (task training + legal training).
6. .
7. Business is responsible for coming up with a process to come up with the Client Care Plan.
8. .
9. Business must ensure all employee records, client records, care plans, billing information, etc. are accurate and kept in a secure location.
10. .
11. Business must establish a clear communication plan between PCPs and the management team.
12. .
13. Business is responsible for training new hires on how to use the app, we are not responsible for this training.

3.4 ROADMAP

1. Gather requirements: Meet with the business to discuss their requirements for the home health care application. Identify the tasks that fall within the standard list and those that require additional fees and what those fees will be.
2. Design the application: Create a high-level design of the application (flowchart).
3. .
4. Develop the task list management section for additional fee tasks: Implement the section responsible for managing the list of tasks, including the standard fee tasks. This section will allow for easy modification of the tasks.
5. .
6. Develop the billing and invoicing section: Implement the section responsible for generating invoices to clients based on the tasks performed by the PCP.
7. .
8. Test the application: Test the application to ensure that it functions as expected and meets the requirements specified by the business.
9. .
10. Maintain the application: Monitor the application for bugs and issues and make updates and modifications as necessary to ensure that it continues to meet the business' needs.

4. DESIGN (Jason Junior)

The proposed solution necessitates the procurement of tablet devices capable of running the program by the business entity, in a quantity deemed necessary for optimal usage by its employees. Our software will be installed on each of the devices, and our team will provide training sessions to the management and presently employed PCPs on how to effectively operate the application.

Paragraph #2

The program is being designed in separate portions for the billing of Primary Services, billing of Secondary/Additional Services, and the authorization from the client to bill for these services.

Paragraph #4

4.1 MODEL OF DESIGN / PROTOTYPING

The Model of Design used to create this program is not a prototyping model since the program was created in one session and fully meets the *Software requirements* provided by the client.

4.1/4.2 APPLICATION FLOWCHART / DESIGN MODEL

Link to flowchart

5. IMPLEMENTATION (Frederico Freshman)

Please see attached .txt file. We used Microsoft Visual Studio Code to write and run the program. The program also successfully runs with the online compiler Programiz. For an overview of the program please see the attached documentation page in the next section.

Program Code – saved as a .py file and submitted with this documentation

Example of Leading Commentary

“””

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This Home Health Care program was written by F. Freshman and J. Junior (Group 3)
Final Project for CIS 245 -Spring XXXX

The Program specifications
Our client wishes to provide their Care Providers and Clients with software that will allow them to calculate billing and authorize the charges for the Care Provider and Client respectively. Our client requests a software solution from us and has given us the ability to acquire hardware and issue training relevant to the process of using this software for business purposes.

Group Members – S. Sophomore, J. Junior, F. Freshman

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Program Output – saved as a .jpg file and submitted with this documentation

6. DOCUMENTATION, MAINTENANCE, AND SUPPORT (Jason Junior & Frederico Freshman)

If an issue pertaining to the program arises our team will have options for our clients to contact us for support. Our window of support extends five years after the acceptance of the program and includes support in releasing updates to accommodate new needs related to the program, potentially releasing a bug fix if necessary, and providing communicated support for usage and documentation of this program.

To contact our team for software support, to request a bug fix, or to request a specific update please send an email or letter to:

FSJ SOFTWARE SOLUTIONSCOMPANY LLC

30 E Lake St, Chicago, IL 60601

For documentation, please see this page: ***Here***

FLOWCHART (EXAMPLE)

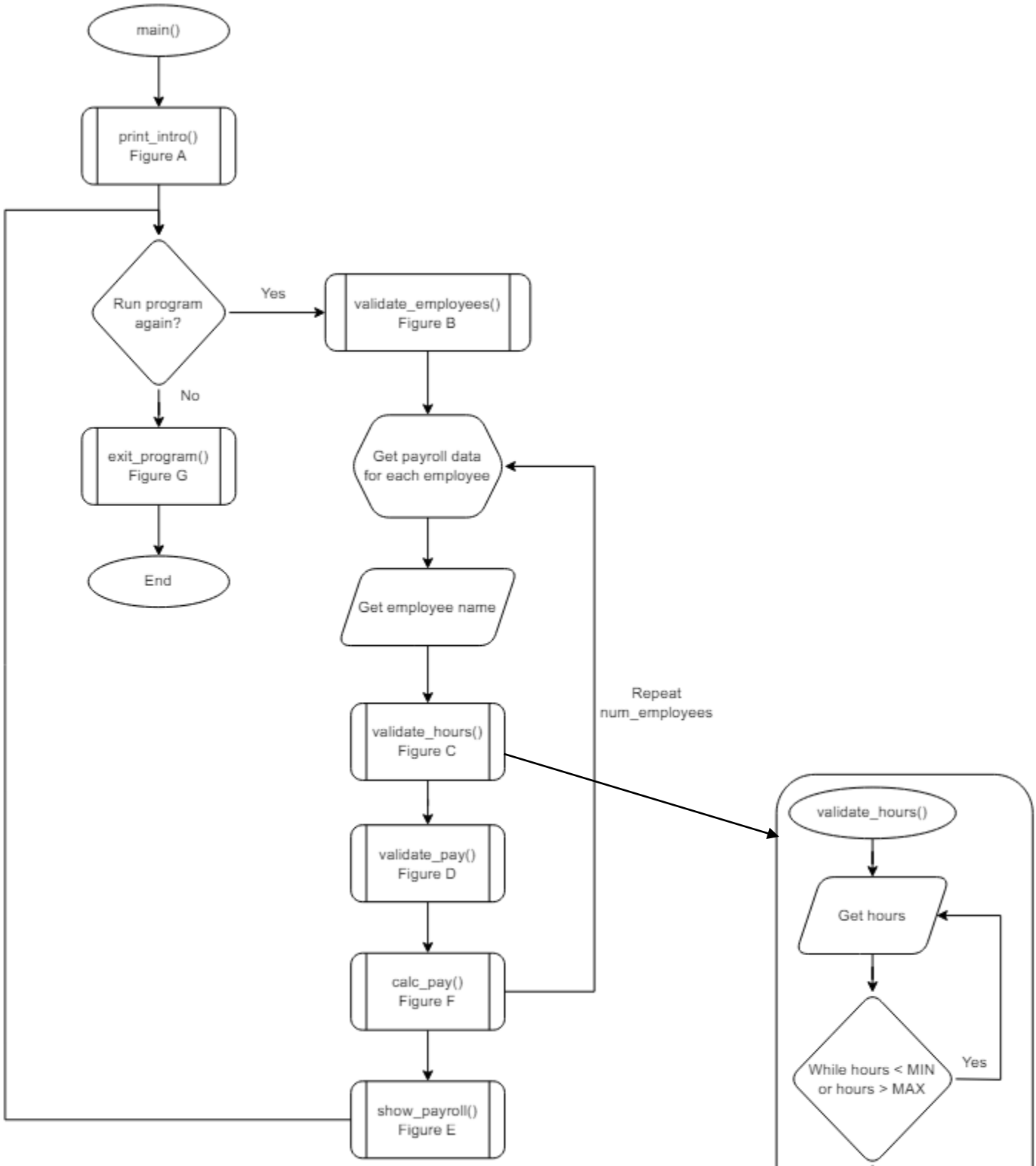


Figure C